

# Mathematics Advanced 11–12 (2024)

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# Mathematics Advanced 11–12 (2024)

## Implementation from 2026

The new Mathematics Advanced 11–12 Syllabus (2024) is to be implemented from 2026 and will replace the [Mathematics Advanced Stage 6 Syllabus \(2017\)](#).

### 2026, Term 1

- Start teaching the new syllabus for Year 11
- Start implementing the new Year 11 school-based assessment requirements
- Continue to teach the [Mathematics Advanced Stage 6 Syllabus \(2017\)](#) for Year 12

### 2026, Term 4

- Start teaching the new syllabus for Year 12
- Start implementing the new Year 12 school-based assessment requirements

### 2027

- First HSC examination for the new syllabus

## Overview

### Syllabus overview

#### Organisation of Mathematics Advanced 11–12

The organisation of outcomes and content for Mathematics Advanced 11–12 highlights the important role Working mathematically plays across all areas of mathematics and reflects the strengthened connections between concepts. Working mathematically has been embedded in the outcomes and content of the syllabus.

Mathematics Advanced outcomes and their related content are organised into 7 areas of study:

- Functions
- Trigonometric functions
- Sequences and series
- Calculus
- Exponential and logarithmic functions
- Statistical analysis
- Financial mathematics

Figure 1 shows the organisation of Mathematics Advanced 11–12, including the focus areas to be completed in each area of study.

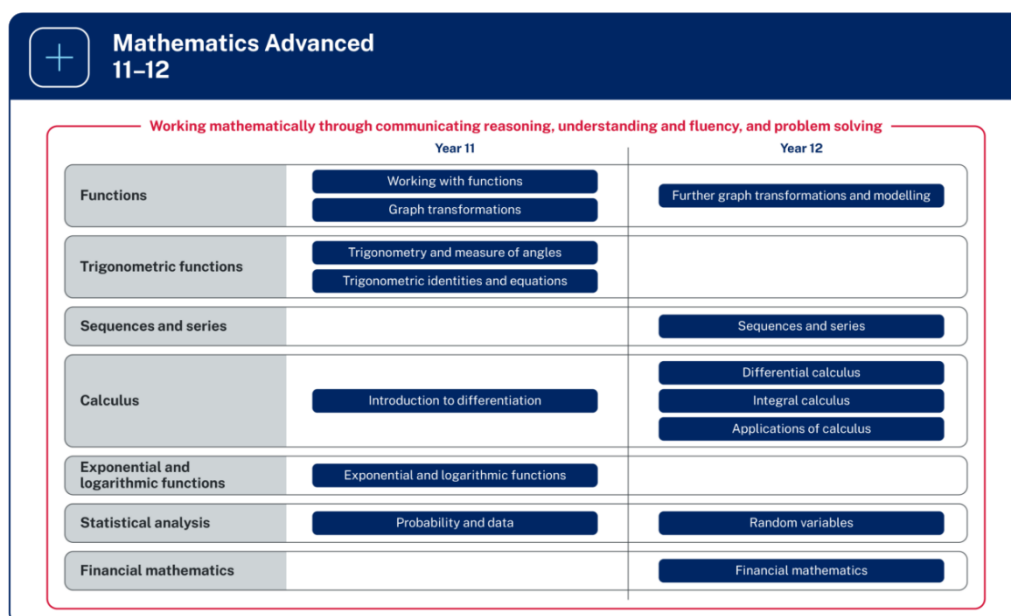


Figure 1: Organisation of Mathematics Advanced 11–12

*Image long description:* Seven rows represent the areas of study for Mathematics Advanced 11–12. Each row is divided in half, showing the focus areas for Year 11 and Year 12. The Functions area of study includes Year 11 focus areas Working with functions and Graph transformations and the Year 12 focus area Further graph transformations and modelling. The Trigonometric functions area of study includes the Year 11 focus areas Trigonometry and measure of angles and Trigonometric identities and equations. The Exponential and logarithmic function area of study includes the Year 11 focus area Exponential and logarithmic functions. The Sequences and series area of study includes the Year 12 focus area Sequences and series. The Calculus area of study includes the Year 11 focus area Introduction to differentiation, and the Year 12 focus areas Differential calculus, Integral calculus and Applications of calculus. Finally, the Statistical analysis area of study includes the Year 11 focus area Probability and data, and the Year 12 focus area Random variables. The Financial mathematics area of study includes the Year 12 focus area Financial mathematics. All content is surrounded by a box labelled with the phrase, ‘Working mathematically through communicating reasoning, understanding and fluency, and problem solving’.

## Protocols for collaborating with Aboriginal and Torres Strait Islander Communities

NESA is committed to working in partnership with Aboriginal Communities and supporting teachers, schools and schooling sectors to improve educational outcomes for young people.

It is important to respect appropriate ways of interacting with Aboriginal Communities and Cultural material when teachers plan, program and implement learning experiences that focus on Aboriginal and Torres Strait Islander Priorities.

Indigenous Cultural and Intellectual Property (ICIP) protocols need to be followed. Aboriginal and Torres Strait Islander Peoples’ ICIP protocols include Cultural Knowledges, Cultural Expression and Cultural Property and documentation of Aboriginal and Torres Strait Islander Peoples’ Identities and lived experiences. It is important to recognise the diversity and complexity of different Cultural groups in NSW, as protocols may differ between local Aboriginal Communities.

Teachers should work in partnership with Elders, parents, Community members, Cultural Knowledge Holders, or a local, regional or state Aboriginal Education Consultative Group. It is important to respect Elders and the roles of men and women. Local Aboriginal Peoples should be invited to share their Cultural Knowledges with students and staff when engaging with Aboriginal histories and Cultural Practices.

## Course description

Mathematics Advanced 11–12 focuses on mathematical ways of viewing the world to investigate concepts, such as order, relation, pattern, uncertainty and generality. The course provides students with the opportunity to explore mathematical problems through observation, reflection and reasoning.

### What students learn

Through the study of Mathematics Advanced 11–12, students:

- develop knowledge, understanding and skills in Working mathematically and communicating concisely and precisely
- consider various applications of mathematics in a broad range of contemporary contexts through mathematical modelling
- gain an appropriate mathematical background for future pathways which involve mathematics and its applications at the tertiary level.

## Course structure and requirements

### Year 11 course structure

| Area of study                         | Focus areas                                                                  |
|---------------------------------------|------------------------------------------------------------------------------|
| Functions                             | Working with functions<br>Graph transformations                              |
| Trigonometric functions               | Trigonometry and measure of angles<br>Trigonometric identities and equations |
| Calculus                              | Introduction to differentiation                                              |
| Exponential and logarithmic functions | Exponential and logarithmic functions                                        |
| Statistical analysis                  | Probability and data                                                         |

## Year 12 course structure

| Area of study         | Focus area                                                             |
|-----------------------|------------------------------------------------------------------------|
| Functions             | Further graph transformations and modelling                            |
| Calculus              | Differential calculus<br>Integral calculus<br>Applications of calculus |
| Sequences and series  | Sequences and series                                                   |
| Statistical analysis  | Random variables                                                       |
| Financial mathematics | Financial mathematics                                                  |

## Safety and risk management

Schools are required to ensure they follow [safety and risk management](#) in delivering the *Mathematics Advanced 11–12 Syllabus*.

## Course enrolment details

### Mathematics Advanced Year 11

- Course number: 11255
- Course hours: 120
- Course units: 2
- Enrolment type: Elective
- Endorsement type: Board developed

### Exclusions

- Mathematics Standard (Year 11, 2 units): 11236
- Mathematics Life Skills (Year 11, 2 units): 17680

### Prerequisites

- Nil

### Corequisites

- Nil

### Mathematics Advanced Year 12

- Course number: 15255
- Course hours: 120
- Course units: 2
- Enrolment type: Elective
- Endorsement type: Board developed

### Exclusions

- Mathematics Standard 1 (Year 12, 2 units): 15231
- Mathematics Standard 1 (examination) (Year 12, 2 units): 15232

- Mathematics Standard 2 (Year 12, 2 units): 15236
- Mathematics Life Skills (Year 12, 2 units): 17680

**Prerequisites**

- Mathematics Advanced (Year 11, 2 units): 11255

**Corequisites**

- Nil

**HSC information**

Information about [curriculum requirements](#) for the HSC are available on [Assessment Certification Examination \(ACE\)](#).

# Rationale

## Mathematics in the Stage 6 Curriculum

Mathematical ideas have evolved and continue to develop across cultures and have been practised in Australia by Aboriginal and Torres Strait Islander Peoples for thousands of years.

The utility of mathematics extends from counting and measuring to explaining real-world phenomena and its inherent beauty. It has evolved in sophisticated ways to become the language used to describe and understand much of the modern world.

The creative application of mathematical concepts is integral to scientific and technological advancement. The symbolic nature of mathematics provides a precise means of communication.

The Mathematics Stage 6 syllabuses are designed to offer students opportunities to think mathematically. Mathematical thinking is supported through questioning, communicating, reasoning, reflecting and the use of appropriate technology, and is encouraged through providing students with opportunities to generalise, challenge, find connections and to think critically and creatively. Using mathematical thinking, students apply their knowledge and skills to deepen their understanding of the world.

## Mathematics Advanced

Mathematics Advanced, Mathematics Extension 1 and Mathematics Extension 2 form a continuum to provide students with opportunities to acquire knowledge, understanding and skills in mathematical concepts at progressively higher levels and with applications in an increasing number of contexts. Mathematics Advanced, Mathematics Extension 1 and Mathematics Extension 2 provide students with opportunities to learn about the interconnected nature of mathematics, its beauty and its functionality. The concepts and techniques of differential and integral calculus are the basis of the courses.

The *Mathematics Advanced 11–12 Syllabus (2024)* is designed to encourage students to appreciate mathematical ways of viewing the world to investigate concepts, such as order, relation, pattern, uncertainty and generality. The course provides students with the opportunity to explore mathematical problems through observation, reflection and reasoning.

The *Mathematics Advanced 11–12 Syllabus (2024)* enables students to use mathematical models and serves as a basis for further studies at the tertiary level in science and commerce that require mathematics and its applications.

## Aim

The aim of Mathematics Advanced in Years 11 and 12 is to enable students to enhance their knowledge and understanding from Stage 5 of how to work mathematically, make mathematical connections, develop their understanding of the relationship between real-world problems and mathematical models, and extend their skills to apply the language of mathematics to communicate in a concise and systematic manner.

## Table of outcomes

### MAO-WM-01 Working mathematically

develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly

This outcome is aligned to all content in each Stage.

| Year 11                                                                                                                         | Year 12                                                                                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>MAV-11-01</b><br>applies algebraic techniques and the laws of indices and surds to manipulate expressions and solve problems | <b>MAV-12-01</b><br>uses algebraic and graphical techniques to analyse trigonometric functions                                                    |
| <b>MAV-11-02</b><br>uses functions and relations to model, analyse and solve problems                                           | <b>MAV-12-02</b><br>models, analyses and solves practical problems involving functions and their transformations                                  |
| <b>MAV-11-03</b><br>analyses and solves algebraic and graphical problems involving transformations of functions and relations   | <b>MAV-12-03</b><br>uses arithmetic and geometric sequences and series to model and solve problems                                                |
| <b>MAV-11-04</b><br>applies trigonometry to solve problems involving geometric shapes                                           | <b>MAV-12-04</b><br>selects and applies differentiation methods to solve problems                                                                 |
| <b>MAV-11-05</b><br>uses periodic functions to solve trigonometric equations and prove trigonometric identities                 | <b>MAV-12-05</b><br>solves problems involving indefinite and definite integrals                                                                   |
| <b>MAV-11-06</b><br>interprets the meaning of the derivative and determines the derivative of functions to solve problems       | <b>MAV-12-06</b><br>applies calculus to graph functions and model and solve problems involving optimisation, rates of change and motion in a line |
| <b>MAV-11-07</b><br>applies exponential and logarithmic laws to simplify expressions, solve equations and prove results         | <b>MAV-12-07</b><br>solves problems involving discrete probability distributions, continuous random variables and the normal distribution         |
| <b>MAV-11-08</b><br>analyses graphs of exponential and logarithmic functions                                                    | <b>MAV-12-08</b><br>models and solves problems to make informed decisions about financial situations                                              |
| <b>MAV-11-09</b><br>solves problems involving probability in a variety of contexts                                              |                                                                                                                                                   |
| <b>MAV-11-10</b><br>displays and analyses datasets using summary statistics and graphical representations                       |                                                                                                                                                   |

# Outcomes and content for Year 11

## Working with functions

### Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- applies algebraic techniques and the laws of indices and surds to manipulate expressions and solve problems **MAV-11-01**
- uses functions and relations to model, analyse and solve problems **MAV-11-02**

### Content

#### Algebraic techniques

- Use [index](#) laws to simplify [expressions](#) and solve problems involving positive, negative, [zero](#) or fractional [indices](#)
- Expand, [factorise](#) and simplify [algebraic expressions](#)
- Simplify expressions involving [algebraic fractions](#)
- Expand and simplify expressions involving [surds](#)
- Identify a [conjugate](#) of  $\sqrt{a} \pm \sqrt{b}$ , and rationalise the [denominators](#) of expressions of the form  $\frac{a}{\sqrt{b} \pm \sqrt{c}}$  and  $\frac{\sqrt{a} \pm \sqrt{d}}{\sqrt{b} \pm \sqrt{c}}$ , where  $a, b, c$  and  $d$  are positive [rational numbers](#)
- Solve [quadratic equations](#)  $ax^2 + bx + c = 0$  by factorisation, [completing the square](#) and using the [quadratic formula](#)  $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$  where  $a, b$  and  $c$  are [real numbers](#) and  $a \neq 0$
- Define the [discriminant](#) as  $\Delta = b^2 - 4ac$  and use it to solve problems involving the number of real [roots](#) of a quadratic equation and determine the conditions for roots to be [equal](#), [distinct](#), real or rational

#### Introduction to functions and relations

- Describe a [relation](#) between two [sets](#) as an [association](#) between the [elements](#) of one set and the elements of the other set
- Define a real [function](#)  $f$  of a real [variable](#)  $x$  as a relation where each element  $x$  of a given set is associated with exactly one element  $y$  of the second set
- Recognise that a relation is a rule which can be represented by an algebraic formula, a table of values, a set of ordered pairs  $(x, y)$  or a [graph](#)
- Use the notation  $f(x)$  to identify the unique value of  $y$  associated with  $x$  when working with functions, and refer to it as the value of  $f$  at  $x$
- Refer to  $x$  in a function  $y = f(x)$  as the [independent variable](#) of the function, and refer to  $y$  as the [dependent variable](#)
- Substitute numeric and algebraic expressions into the formulas of functions
- Apply the [vertical line test](#) on the graph of a relation to determine whether it represents a function
- Define the [domain](#) of a function  $f$  as the set of real numbers on which  $f$  is defined
- Define the [range](#) of a function  $f$  as the set of values of  $f(x)$  obtained as  $x$  varies over the domain of  $f$

- Refer to a value in the domain of a function at which the function is 0 as a zero of the function
- Recognise that the  $x$  -[intercepts](#) of the graph of  $y = f(x)$  are its zeroes, the solutions of  $f(x) = 0$ , and that the  $y$  -intercept is  $f(0)$

### Linear functions

- Determine the [equations](#) of straight lines in [gradient–intercept form](#)  $y = mx + c$  with [gradient](#)  $m$  and  $y$  -intercept  $c$
- Determine the equations of straight lines in [general form](#)  $ax + by + c = 0$ , where  $a$ ,  $b$  and  $c$  are [constants](#)
- Determine the equation of a straight line passing through a point  $(x_1, y_1)$  with gradient  $m$  using the point–gradient formula  $y - y_1 = m(x - x_1)$
- Determine the equation of a straight line passing through two points  $(x_1, y_1)$  and  $(x_2, y_2)$  by calculating its gradient  $m$  using the formula  $m = \frac{y_2 - y_1}{x_2 - x_1}$
- Determine the  $x$  -intercept and  $y$  -intercept of a straight line given its equation
- Choose and apply appropriate techniques to graph a straight line given its equation
- Find the equation of a line that is [parallel](#) or [perpendicular](#) to a given line
- Solve [linear inequalities](#) and graph the solution on a number line

### Quadratic and cubic functions

- Identify the  $x$  -intercepts of a [parabola](#) whose [quadratic function](#) is expressed in factored form
- Use the discriminant to determine the number of  $x$  -intercepts on a parabola and justify its position in relation to the  $x$  -axis
- Show by completing the square on the general quadratic  $y = ax^2 + bx + c$  that the [axis of symmetry](#) is  $x = -\frac{b}{2a}$  where  $a$ ,  $b$  and  $c$  are constants
- Identify the axis of symmetry and [vertex](#) of a parabola by completing the square on its quadratic function
- Choose and apply appropriate techniques to graph a parabola of the form  $y = ax^2 + bx + c$  by identifying its  $x$  -intercepts if they exist, its  $y$  -intercept, its axis of symmetry using  $x = -\frac{b}{2a}$  and its vertex
- Find the equation of a parabola given sufficient graphical features
- Use the fact that two quadratic functions are equal for all values of  $x$  if and only if the corresponding [coefficients](#) are equal to solve related problems
- Recognise that solving  $f(x) = k$  for some constant  $k$  corresponds to finding the  $x$  -coordinate(s) of the [intersection](#) of the graphs  $y = f(x)$  and  $y = k$
- Solve problems by finding the solution to simultaneous equations involving a [linear](#) and a quadratic function, or two quadratic functions, both algebraically and graphically
- Solve [quadratic inequalities](#)
- Recognise and graph [cubic functions](#) of the form  $f(x) = kx^3$  and  $f(x) = k(x - a)(x - b)(x - c)$ , where  $a$ ,  $b$ ,  $c$  and  $k$  are constants and  $k \neq 0$

### Reciprocal functions

- Graph functions of the form  $f(x) = \frac{k}{x}$ , where  $k$  is a constant and  $k \neq 0$ , and identify their hyperbolic shape and their [asymptotes](#)
- Describe the behaviour of  $f(x) = \frac{k}{x}$  as  $x \rightarrow \infty$  and  $x \rightarrow -\infty$

## Constructing and using functions

- Construct and use linear functions to model and solve problems in real-world situations, identifying the independent and dependent variables and any restrictions on these variables, and justify conclusions in the context of the problem
- Use linear inequalities to model and solve problems in real-world situations, and justify conclusions in the context of the problem
- Solve practical problems involving a pair of simultaneous [linear equations](#) both algebraically and graphically, with and without graphing applications, and justify conclusions in the context of the problem
- Construct and use simultaneous equations to model and solve a problem where [cost](#) and [revenue](#) are represented by linear equations, identify and analyse the [break-even point](#), and justify conclusions in the context of the problem
- Model and solve practical problems involving quadratic functions and justify conclusions in the context of the problem

## Direct and inverse variation

- Develop [models](#) of the form  $y = kx^n$ , where  $k$  is a non-zero constant, from descriptions of situations in which one quantity varies directly with another
- Develop the model  $y = \frac{k}{x^n}$ , where  $k$  is a non-zero constant, from descriptions of situations in which one quantity varies inversely with another
- Evaluate  $k$  in the equations  $y = kx^n$  and  $y = \frac{k}{x^n}$ , given one pair of values for the variables, and use the resulting formula to find other values of the variables
- Analyse and solve problems involving [direct](#) and [inverse variation](#)

## Circles and semicircles

- Derive the equation of a [circle](#) of radius  $r$  with centre at the origin by considering [Pythagoras' theorem](#)
- Graph circles of the form  $x^2 + y^2 = r^2$  from their equations
- Determine the equation of a circle of the form  $x^2 + y^2 = r^2$  given its graph
- Identify and graph the [semicircles](#)  $y = \sqrt{r^2 - x^2}$ ,  $y = -\sqrt{r^2 - x^2}$ ,  $x = \sqrt{r^2 - y^2}$  and  $x = -\sqrt{r^2 - y^2}$

## Properties of functions, relations and graphs

- Extend the definitions of domain and range to relations
- Recognise domains and ranges of functions and relations given in [interval notation](#), as inequalities and as worded descriptions
- Determine and describe the domain and range of functions and relations, using interval notation, inequalities or worded descriptions
- Define a function to be even if its graph is unchanged under [reflection](#) in the  $y$ -axis, and [odd](#) if its graph is unchanged under [rotation](#) of  $180^\circ$  about the origin
- Develop and use the tests that a function  $f(x)$  is odd if  $f(-x) = -f(x)$  and a function  $f(x)$  is even if  $f(-x) = f(x)$
- Solve problems involving even and odd functions
- Use the [composite function](#)  $f(g(x))$ , where the output of  $g(x)$  becomes the input of  $f(x)$
- Determine the equations of composite functions

## Piecewise-defined functions

- Interpret [piecewise-defined functions](#), where the function is defined differently in different parts of the domain
- Graph piecewise-defined functions involving functions covered in the scope of the Mathematics Advanced course, test if they are even or odd, and determine the domain and range
- Define informally that a function is continuous at a point if the curve can be drawn through the point without lifting the pen off the paper
- Identify points where piecewise-defined functions and other functions are not continuous
- Define a [discontinuity](#) of a function informally as a point where the function is not continuous

## Absolute value functions

- Define the [absolute value](#)  $|x|$  of a number  $x$ , also known as the [magnitude](#) of  $x$ , to be the [distance](#) from the origin to  $x$  on the number line
- Establish and use the piecewise definition  $|x| = \begin{cases} x, & \text{for } x \geq 0, \\ -x, & \text{for } x < 0 \end{cases}$
- Show using numerical [substitutions](#) that  $\sqrt{x^2} = |x|$  and use the result
- Graph the function  $y = |x|$ , describe its symmetry, and identify its domain and range
- Graph  $y = |ax + b|$  with and without graphing applications, and identify its symmetry, domain and range
- Solve absolute value equations of the form  $|ax + b| = k$  algebraically and graphically

# Trigonometry and measure of angles

## Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- applies trigonometry to solve problems involving geometric shapes **MAV-11-04**

## Content

### Trigonometry with acute angles

- Use [Pythagoras' theorem](#) to find the exact [sine](#), cosine and [tangent ratios](#) for [angles](#) of  $30^\circ$ ,  $45^\circ$  and  $60^\circ$
- Apply trigonometry to solve problems involving right-angled triangles in two dimensions, [true](#) and [compass bearings](#), and [angles of elevation](#) and [depression](#)

### Trigonometry with angles of any magnitude

- Represent angles of any [magnitude](#) using rays from the origin in the [Cartesian plane](#) and describe how one ray represents [infinitely](#) many angles
- Develop the definitions  $\cos \theta = \frac{x}{r}$ ,  $\sin \theta = \frac{y}{r}$  and  $\tan \theta = \frac{y}{x}$ , where  $(x,y)$  is a point on the [circle](#) of radius  $r$ , centred at the origin, and  $\theta$  is the angle between the positive  $x$ -axis and the radius drawn to this point
- Use the definitions  $\cos \theta = \frac{x}{r}$ ,  $\sin \theta = \frac{y}{r}$  and  $\tan \theta = \frac{y}{x}$  to evaluate  $\sin \theta$ ,  $\cos \theta$  and  $\tan \theta$  where  $\theta$  is a multiple of  $90^\circ$ , and identify the values of  $\theta$  for which these ratios are undefined
- Extend and apply the definitions  $\cos \theta = \frac{x}{r}$ ,  $\sin \theta = \frac{y}{r}$  and  $\tan \theta = \frac{y}{x}$  for angles of any magnitude
- Identify the related angle of an angle of any magnitude, excluding multiples of  $90^\circ$ , as the [acute angle](#) between the ray and the  $x$ -axis and obtain the values of the trigonometric [functions](#) of an angle of any magnitude from the trigonometric functions of the related angle
- Develop and use the [trigonometric ratios](#) for angles that can be written in the form  $\theta = 180^\circ \pm A$ , and  $\theta = 360^\circ - A$ , where  $0^\circ < A < 90^\circ$
- Establish and use the results  $\cos(-\theta) = \cos \theta$ ,  $\sin(-\theta) = -\sin \theta$  and  $\tan(-\theta) = -\tan \theta$
- Examine the [proof](#) of the [sine rule](#)  $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$ , [cosine rule](#)  $c^2 = a^2 + b^2 - 2ab \cos C$  and the [area of a triangle](#) formula  $A = \frac{1}{2}ab \sin C$  for a given triangle  $ABC$
- Use graphing applications or geometric construction to examine the [ambiguous case of the sine rule](#), in which there are two possible solutions for an angle, and the condition for it to arise
- Apply the sine rule, cosine rule and formula for the area of a triangle to solve problems where angles are measured in [degrees](#), or degrees and [minutes](#)

## Radians

- Recognise that both ratios  $\frac{\text{arc length}}{\text{circumference}}$  and  $\frac{\text{area of sector}}{\text{area of circle}}$  are [equal](#) to  $\frac{\theta}{\text{one revolution}}$  where  $\theta$  is the angle at the centre of a circle [subtended](#) by the [arc](#)
- Define the angle size of  $\theta$  in [radian measure](#) as the ratio  $\frac{\text{arc length}}{\text{radius of a circle}}$
- Explain why  $360^\circ = 2\pi$  [radians](#) and  $1 \text{ radian} = \frac{180}{\pi}$  degrees

- Convert between degrees and radians and find the exact sine, cosine and tangent ratios for [integer](#) multiples of  $\frac{\pi}{6}$  and  $\frac{\pi}{4}$
- Graph  $y = \sin x$ ,  $y = \cos x$  and  $y = \tan x$  over [domains](#) given in degrees or radians, showing [intercepts](#) with the  $x$ -axis and  $y$ -axis and any [asymptotes](#), determine their domains and [ranges](#), and [period](#) and [amplitude](#) where appropriate, and whether each is even or [odd](#) or neither
- Establish and use the formula  $l = r\theta$  for the length  $l$  of arc subtending an angle  $\theta$  in radians at the centre of a circle of radius  $r$
- Prove and use the formula  $A = \frac{1}{2}r^2\theta$  for the area of a [sector](#) with angle  $\theta$  in radians at the centre of a circle of radius  $r$
- Solve problems involving [arc lengths](#) and areas of major and minor sectors and [segments](#)

# Trigonometric identities and equations

## Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- uses periodic functions to solve trigonometric equations and prove trigonometric identities **MAV-11-05**

## Content

- Define the trigonometric [functions](#)  $\sec \theta$ ,  $\operatorname{cosec} \theta$  and  $\cot \theta$  for [acute angles](#)  $\theta$  using ratios of sides in a right-angled triangle
- Find the exact [secant](#), [cosecant](#) and [cotangent ratios](#) for [angles](#) of  $30^\circ$ ,  $45^\circ$  and  $60^\circ$
- Justify the values of  $\sec \theta$ ,  $\operatorname{cosec} \theta$  and  $\cot \theta$  at  $0^\circ$  and  $90^\circ$  by examining the ratios of sides in a right-angled triangle as  $\theta$  tends to  $0^\circ$  and  $90^\circ$
- Develop the definitions  $\sec \theta = \frac{r}{x}$ ,  $\operatorname{cosec} \theta = \frac{r}{y}$  and  $\cot \theta = \frac{x}{y}$  using a [circle](#) of radius  $r$  centred at the origin
- Establish the [reciprocal](#) ratios  $\sec A = \frac{1}{\cos A}$ ,  $\operatorname{cosec} A = \frac{1}{\sin A}$ , and  $\cot A = \frac{1}{\tan A}$ , identifying the angles to be excluded in each [identity](#)
- Determine the [exact value](#) of the secant, cosecant and cotangent ratios for angles that are [integer](#) multiples of  $\frac{\pi}{6}$  and  $\frac{\pi}{4}$ , if they exist
- Establish the identities  $\tan \theta = \frac{\sin \theta}{\cos \theta}$  and  $\cot \theta = \frac{\cos \theta}{\sin \theta}$ , identifying the angles to be excluded in each identity
- Prove the [complementary angle](#) identities  $\sin(90^\circ - \theta) = \cos \theta$ ,  $\cos(90^\circ - \theta) = \sin \theta$ ,  $\tan(90^\circ - \theta) = \cot \theta$ ,  $\cot(90^\circ - \theta) = \tan \theta$ ,  $\sec(90^\circ - \theta) = \operatorname{cosec} \theta$  and  $\operatorname{cosec}(90^\circ - \theta) = \sec \theta$  identifying the angles to be excluded in each identity
- Evaluate trigonometric [expressions](#) using angles of any [magnitude](#) and complementary angle identities
- Solve [equations](#) involving [trigonometric ratios](#) of angles, specified in [degrees](#) or [radians](#), on a restricted [domain](#)
- Prove the Pythagorean identity  $\cos^2 x + \sin^2 x = 1$ , and the identities  $1 + \tan^2 x = \sec^2 x$  and  $1 + \cot^2 x = \operatorname{cosec}^2 x$
- Apply trigonometric identities to solve problems, simplify expressions and prove further trigonometric identities using [substitution](#) and/or [reduction](#) to  $\sin x$  and  $\cos x$
- Solve problems involving trigonometric equations, including those that reduce to [quadratic equations](#), on a restricted domain

# Introduction to differentiation

## Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- interprets the meaning of the derivative and determines the derivative of functions to solve problems **MAV-11-06**

## Content

### Estimating change

- Define the [average rate of change](#) of  $y$  with respect to  $x$  for a [function](#)  $y = f(x)$  over the [domain](#)  $[a, b]$  as  $\frac{\Delta y}{\Delta x} = \frac{\text{change in } y}{\text{change in } x}$ , that is  $\frac{\Delta y}{\Delta x} = \frac{f(b) - f(a)}{b - a}$ , and recognise  $\frac{f(b) - f(a)}{b - a}$  as the [gradient](#) of the [secant](#) through  $(a, f(a))$  and  $(b, f(b))$  on the [graph](#) of  $y = f(x)$
- Recognise [speed](#) as a [rate of change](#) of [distance](#) with respect to time
- Use the definition for average rate of change to determine the [average](#) speed of an object from a given [distance–time graph](#)
- Describe the difference between the average speed of an object and its instantaneous speed
- Determine that the instantaneous speed of an object at time  $t$  can be approximated by the average speed between its position at time  $t$  and its position some time later, and explain how this approximation can be improved
- Relate the instantaneous speed of an object to the gradient of the [tangent](#) at that point on its distance–time graph
- Estimate the instantaneous speed of an object from its distance–time graph
- Recognise when modelling with a [linear function](#) that its gradient is the rate of change and determine the rate of change for linear functions in practical situations
- Recognise when modelling with a non-linear function that the rate of change is not [constant](#) and is represented by the gradient of the tangent to the curve at each point on the curve
- Estimate the [instantaneous rate of change](#) of a non-linear function at a given point from a given graph of a practical situation

### The derivative

- Examine the gradient of a curve at a point on the curve using graphing applications
- Approximate the gradient of a curve  $f(x) = x^n$  at a point  $P(c, f(c))$  by considering the gradient of the secant through  $P$  and  $Q(c + h, f(c + h))$  as the [magnitude](#) of  $h$  approaches [zero](#), using graphing applications or a spreadsheet
- Infer that  $nx^{n-1}$  is the gradient of  $f(x) = x^n$  and verify the result using graphing applications
- Define  $f'(x)$ , for any function  $f(x)$  and any value  $x$ , to be the gradient of the tangent to the curve  $y = f(x)$  at the point  $P(x, f(x))$  if the tangent exists and is not vertical
- Refer to  $f'(x)$  as the [derivative](#) of  $f(x)$  or the gradient, or derived, function of  $f(x)$
- Define [differentiation](#) as the process of finding the derivative of a function
- Find derivatives of constant and linear functions

- Define the derivative of the function  $f(x)$  from first principles, as the limiting value of the gradient of the secant  $\frac{f(x+h)-f(x)}{h}$  as  $h$  approaches zero, when this limiting value exists, and use the notation  $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h)-f(x)}{h}$
- Use first principles to find the derivative of [quadratic functions](#)

### Calculations with the derivative

- Use the notation  $\frac{dy}{dx}$  and  $y'$  for the derivative of  $y$  when  $y$  is a function of  $x$
- Use the notation  $\frac{d}{dx}(f(x))$  and  $f'(x)$  for the derivative of a function  $f(x)$
- Use the formula  $\frac{d}{dx}(x^n) = nx^{n-1}$  for all real values of  $n$
- Apply the fact that the derivative of a sum is the sum of the derivatives:  
 $\frac{d}{dx}(f(x) + g(x)) = f'(x) + g'(x)$ , and the derivative of a multiple of a function is the multiple of its derivative:  $\frac{d}{dx}(k \cdot f(x)) = k \cdot f'(x)$
- Use the rules for differentiation to find [equations](#) of tangents and [normals to a curve](#) at points on the curve
- Find points on a curve where the tangent or normal has a given gradient
- Examine and use the relationship between the [angle of inclination](#) of a line or tangent to a curve,  $\theta$ , with the positive  $x$ -axis, and the gradient,  $m$ , of that line or tangent, and establish that  $\tan \theta = m$
- Apply the [product rule](#): if  $y = uv$ , where  $u$  and  $v$  are both [differentiable](#) functions of  $x$ , then  $\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$  or if  $h(x) = f(x)g(x)$  for differentiable functions  $f(x)$  and  $g(x)$  then  $h'(x) = f(x)g'(x) + f'(x)g(x)$
- Apply the [quotient rule](#): if  $y = \frac{u}{v}$ , where  $u$  and  $v$  are both differentiable functions of  $x$ , then  $\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$  or if  $h(x) = \frac{f(x)}{g(x)}$  then  $h'(x) = \frac{g(x)f'(x) - f(x)g'(x)}{[g(x)]^2}$
- Apply the [chain rule](#): if  $y$  is a differentiable function of  $u$ , and  $u$  is a differentiable function of  $x$ , then  $\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$  or if  $h(x) = f(g(x))$  for differentiable functions  $f(x)$  and  $g(x)$  then  $h'(x) = f'(g(x))g'(x)$
- Identify and apply the product, quotient or chain rule, or a combination of the rules, as appropriate to differentiate a given function

### Graphical applications of the derivative

- Interpret  $f'(x)$  as increasing at  $x = c$  when  $f'(c) > 0$  and decreasing at  $x = c$  when  $f'(c) < 0$
- Describe the behaviour of a function at a point as stationary when the tangent at the point is [parallel](#) to the  $x$ -axis, and recognise that  $f(x)$  is stationary at  $x = c$  when  $f'(c) = 0$
- Graph  $y = f'(x)$  for a given graph of a function  $y = f(x)$
- Numerically estimate the value of the derivative at a point on the graph of a power of  $x$ , with and without the use of digital tools
- Identify [stationary points](#) on the graph of a [cubic function](#), and the values of  $x$  for which the function is increasing and/or decreasing, by first calculating the derivative, and justify conclusions

### The derivative as a rate of change

- Interpret  $f'(c)$  as the instantaneous rate of change of the function  $f(x)$  at  $x = c$
- Define and distinguish between [displacement](#) and distance and between [velocity](#) and speed

- Use graphs of functions and their derivatives, without the use of algebraic techniques, to describe and interpret physical phenomena
- Use the notation  $\frac{dx}{dt}$  or  $\dot{x}$  to represent the velocity of a particle with displacement  $x$  from a point as a function of time  $t$
- Solve problems by determining the velocity of a particle moving in a straight line, given its displacement from a point as a function of time

# Exponential and logarithmic functions

## Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- applies exponential and logarithmic laws to simplify expressions, solve equations and prove results **MAV-11-07**
- analyses graphs of exponential and logarithmic functions **MAV-11-08**

## Content

### Exponential functions

- Graph the [exponential functions](#)  $y = k(a^x)$  and  $y = k(a^{-x})$  for [constants](#)  $a$  and  $k$  where  $a > 0$ ,  $a \neq 1$  and  $k \neq 0$ , and identify its [asymptote](#),  $y$ -[intercept](#), [domain](#) and [range](#)
- Describe the behaviour of  $y = k(a^x)$  and  $y = k(a^{-x})$  as  $x \rightarrow \infty$  and  $x \rightarrow -\infty$
- Examine the [gradient](#) of the [tangent](#) to the curve  $y = a^x$  at its  $y$ -intercept for varying values of  $a$ , and verify using graphing applications that there is a unique number  $e \approx 2.71828182845 \dots$ , such that the gradient of the tangent to  $y = e^x$  at  $x = 0$  is 1, and call this number  $e$  [Euler's number](#)
- Examine the gradient [function](#) of  $y = a^x$  with graphing applications and identify that the gradient function is  $\frac{d}{dx}(a^x) = ka^x$ , for some constant  $k$  depending only on  $a$
- Conclude that Euler's number,  $e$ , is a unique number such that  $\frac{d}{dx}(e^x) = e^x$ , that is,  $e^x$  is its own [derivative](#)

### Logarithmic functions

- Define the [logarithm](#) of a number  $y$ , where  $y > 0$ , to any positive base  $a$  as the [index](#) to which  $a$  is raised to give  $y$
- Use the notation  $\log_a y$  for the logarithm of  $y$  to the base  $a$
- Define the [natural logarithm](#)  $\ln a = \log_e a$
- Use digital tools to determine [rational](#) and [irrational](#) values of exponential and logarithmic [expressions](#)
- Explain that  $y = a^x$  is [equivalent](#) to  $x = \log_a y$  for  $a > 0$  and  $a \neq 1$ , and use the equivalence to solve [equations](#) of the form  $a^x = b$ , for  $a > 0$
- Recognise and use the logarithmic properties:  $\log_a a^x = x$  for all real  $x$ ,  $a^{\log_a x} = x$  where  $x > 0$
- Derive the laws of logarithms from the laws of [indices](#)  $\log_a m + \log_a n = \log_a(mn)$ ,  
 $\log_a m - \log_a n = \log_a\left(\frac{m}{n}\right)$  and  $\log_a(m^n) = n\log_a m$
- Justify the logarithmic results  $\log_a a = 1$ ,  $\log_a 1 = 0$  and  $\log_a\left(\frac{1}{x}\right) = -\log_a x$
- Apply the logarithmic laws and results to simplify expressions, solve equations and prove results, using digital tools where necessary
- Prove the [change of base rule](#)  $\log_a x = \frac{\log_b x}{\log_b a}$  and use it to solve problems
- Graph the [logarithmic function](#)  $y = \log_a x$  for  $a > 0$  and  $a \neq 1$

- Use graphing applications to identify the graphs of  $y = a^x$  and  $y = \log_a x$  as [reflections](#) of each other in the line  $y = x$ , in cases where  $a > 1$  and  $0 < a < 1$

# Graph transformations

## Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- analyses and solves algebraic and graphical problems involving transformations of functions and relations **MAV-11-03**

## Content

- Establish using graphing applications that replacing  $x$  by  $-x$  in the [equation](#) of a [graph](#) corresponds to the [reflection](#) of the graph in the  $y$ -axis, and that replacing  $y$  by  $-y$  corresponds to a reflection in the  $x$ -axis
- Establish using graphing applications that replacing  $x$  by  $x - a$  in the equation of a graph corresponds to a horizontal translation of a graph by  $a$ , that replacing  $y$  by  $y - b$  corresponds to a vertical [translation](#) by  $b$ , and recognise that the sign of  $a$  and  $b$  determines the direction of the translation
- Describe a horizontal [dilation](#) as a stretch from the  $y$ -axis and a vertical dilation as a stretch from the  $x$ -axis
- Establish using graphing applications that replacing  $x$  by  $\frac{x}{k}$  in the equation of a graph corresponds to a horizontal dilation of a graph by a factor of  $k$ , that replacing  $y$  by  $\frac{y}{l}$  corresponds to a vertical dilation by a factor of  $l$ , and determine the effects on the graph when the [magnitude](#) of the dilation factor lies between 0 and 1
- Establish, using graphing applications, replacing  $x$  by  $\frac{x}{k}$  and replacing  $y$  by  $\frac{y}{l}$  in the equation of a graph corresponds to a dilation by a factor of  $k$ , that is, an [enlargement](#) of the graph when  $k > 1$  and a [reduction](#) of the graph when  $0 < k < 1$
- Apply reflections, translations and dilations to [functions](#) within the scope of the Mathematics Advanced course, excluding trigonometric functions, determine the function rule and the graph of the transformed function, the [domain](#) and [range](#) of the transformed graph, any [intercepts](#), [asymptotes](#) and [discontinuities](#) where appropriate, and describe which [transformations](#) have been applied
- Use the principles of translations to identify the centre, radius, domain and range of graphs of [circles](#) in the form  $(x - a)^2 + (y - b)^2 = r^2$ , where  $a$ ,  $b$  and  $r$  are [constants](#)
- Find the centre and radius of circles of the form  $x^2 + y^2 + ax + by + c = 0$ , where  $a$ ,  $b$  and  $c$  are constants, by [completing the squares](#)
- Graph circles given their equations and find the equation of a circle from its graph
- Recognise that the order in which individual transformations are applied to functions is important

# Probability and data

## Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- solves problems involving probability in a variety of contexts **MAV-11-09**
- displays and analyses datasets using summary statistics and graphical representations **MAV-11-10**

## Content

### Sets and set notation

- Define a [set](#) as a collection of objects, called the [elements](#) of the set, and represent a set using notation such as  $\{2, 3, 5, 7\}$  and  $\{0, 2, 4, 6, \dots\}$
- Use the notation  $n(A)$  or  $|A|$  to represent the number of elements in a [finite](#) set  $A$
- Define the [empty set](#) as the set with no elements, denoted in set notation as  $\emptyset$
- Use the notation  $\bar{A}$ ,  $A'$  or  $A^c$  to represent the [complement](#) of a set  $A$  with respect to some [universal set](#)  $U$
- Define  $A$  to be a [subset](#) of  $B$  if all the elements of  $A$  are elements of  $B$
- Define the [intersection](#)  $A \cap B$  of sets  $A$  and  $B$  to be the set of elements that are in  $A$  and in  $B$
- Define the [union](#)  $A \cup B$  of sets  $A$  and  $B$  to be the set of elements that are in  $A$  or in  $B$
- Define sets  $A$  and  $B$  to be [disjoint](#) if  $A \cap B = \emptyset$ , that is, they have no elements in common
- Use [Venn diagrams](#) in practical situations to represent and interpret sets that may intersect in various ways within a universal set
- Establish and use the rule  $|A \cup B| = |A| + |B| - |A \cap B|$

### Probability

- Define an [experiment](#) or a [trial](#) to be any procedure that can be [infinitely](#) repeated and has a well-defined set of possible [outcomes](#) known as the [sample](#) space, denoted  $S$
- Identify an [event](#)  $A$  as a subset  $A$  of the sample space  $S$
- Define the [probability](#) of each outcome to be  $\frac{1}{|S|}$  when all the outcomes are [equally likely](#) and the probability of the event  $A$  to be  $P(A) = \frac{|A|}{|S|}$
- Interpret the notation  $\bar{A}$  to be the event 'A does not occur', interpret  $A \cap B$  to be the event 'A and B both occur', and interpret  $A \cup B$  to be the event 'A or B occurs'
- Use Venn diagrams to represent the relationship between events within the same sample space, including [mutually exclusive events](#), that is, events that as subsets of the sample space are disjoint
- Establish and use the rules  $P(\bar{A}) = 1 - P(A)$  and  $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- Use [arrays](#) and [tree diagrams](#) to determine the outcomes and probabilities for [multistage events](#)

### Conditional probability

- Define [conditional probability](#) as the probability that an event  $A$  occurs given that another event  $B$  has already occurred, and use the notation  $P(A|B)$

- Examine conditional probability by restricting the sample space and event spaces in a Venn diagram, using a [two-way table](#), a tree diagram and other arrays
- Establish that  $P(A|B) = \frac{|A \cap B|}{|B|}$  when all outcomes are equally likely by restricting the sample space and event space, and hence  $P(A|B) = \frac{P(A \cap B)}{P(B)}$ , provided  $|B| \neq 0$
- Use the formulas for  $P(A|B)$  to solve practical problems involving conditional probability
- Define two events to be [independent](#) if the occurrence of one event does not affect the probability that the other event occurs
- Explain that two events  $A$  and  $B$  are independent [means](#)  $P(A|B) = P(A)$  and  $P(B|A) = P(B)$ , and show algebraically that if one of these formulas is true, then the other is also true
- Use the formula  $P(A|B) = \frac{P(A \cap B)}{P(B)}$ , and the test  $P(A|B) = P(A)$  for independence to prove that if two events are independent, then  $P(A \cap B) = P(A) \times P(B)$ , and to prove [conversely](#) that if  $P(A \cap B) = P(A) \times P(B)$ , then  $A$  and  $B$  are independent
- Solve practical problems involving independent events

## Data

- Define a [random variable](#) as a [variable](#) whose value is the outcome of a random experiment
- Compare [discrete random variables](#) with [continuous random variables](#), describe their differences, and give practical examples of each
- Organise finite datasets using a table or a spreadsheet, listing the values, [frequency](#), [relative frequency](#), [cumulative frequency](#), and cumulative relative frequency
- Graph the frequency, relative frequency, and [cumulative frequency histograms](#) and [polygons](#) of datasets, using spreadsheets or graphing applications, and identify the [mode](#) and [median](#) from the graphs, and from tables
- Use the relative frequency to estimate the probability of results in experiments

# Outcomes and content for Year 12

## Further graph transformations and modelling

### Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- uses algebraic and graphical techniques to analyse trigonometric functions **MAV-12-01**
- models, analyses and solves practical problems involving functions and their transformations **MAV-12-02**

### Content

#### Transformations of trigonometric functions

- Use graphing applications to explore [reflections](#), [translations](#) and [dilations](#) of, and confirm that the principles of [transformations](#) hold for, the trigonometric [functions](#)  $f(x) = \sin x$ ,  $f(x) = \cos x$  and  $f(x) = \tan x$
- Apply the principles of transformations to graph transformations of the trigonometric functions  $f(x) = \sin x$ ,  $f(x) = \cos x$  and  $f(x) = \tan x$ , describe which transformations have been applied to  $y = f(x)$  and determine the [domain](#), [range](#), horizontal translation, vertical translation, dilation factor, [period](#) and [amplitude](#) when applicable
- Solve [equations](#) and graphical problems involving transformations of the trigonometric functions  $f(x) = \sin x$ ,  $f(x) = \cos x$  or  $f(x) = \tan x$ , using graphing applications or otherwise, within a specified domain

#### Modelling with functions

- Model and solve problems involving periodic phenomena using transformations of the trigonometric functions  $f(x) = \sin x$ ,  $f(x) = \cos x$  or  $f(x) = \tan x$ , without the use of calculus
- Model and solve practical problems involving functions and their transformations, within the scope of the Mathematics Advanced course, using graphical and algebraic techniques
- Recognise that [logarithmic scales](#) are useful to represent a [variable](#) that ranges over a large [set](#) of values and explain when a logarithmic scale is suitable
- Use logarithmic scales to model and solve problems involving decibels (dB) in acoustics, seismic scales for earthquakes, magnitudes of stars and the pH scale in chemistry

# Sequences and series

## Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- uses arithmetic and geometric sequences and series to model and solve problems **MAV-12-03**

## Content

### Sequences and series

- Define a [sequence](#) as an ordered list of objects
- Use the notation  $a_n$ , where  $n$  is a positive [integer](#), to represent the  $n$ th term of a sequence
- Distinguish between a [finite](#) sequence  $a_1, a_2, \dots, a_m$  that terminates at its  $m$ th term, for some [whole number](#)  $m \geq 1$ , and an [infinite](#) sequence  $a_1, a_2, \dots$  that never terminates
- Define the  $n$ th [partial sum](#)  $S_n$  of a sequence  $a_1, a_2, \dots$  to be the sum of the first  $n$  terms of the sequence:  $S_n = a_1 + a_2 + \dots + a_n$ , for all whole numbers  $n \geq 1$
- Define a [series](#) formally as the sum of the terms of an infinite sequence and use the notation  $a_1 + a_2 + a_3 + \dots$  for the series corresponding to the sequence  $a_1, a_2, a_3, \dots$
- Use [summation notation](#) to represent the sum of terms  $a_i$  to  $a_j$  of a sequence where  $j > i$ ,  
$$\sum_{k=i}^j a_k = a_i + a_{i+1} + a_{i+2} + \dots + a_{j-1} + a_j$$

### Arithmetic sequences and series

- Define a sequence  $a_n$  to be an [arithmetic sequence](#), or [arithmetic progression \(AP\)](#), if every difference  $a_n - a_{n-1}$  of successive terms is the same where  $n \geq 2$ , that is  $a_n - a_{n-1} = d$  for some [constant](#)  $d$  called the [common difference](#)
- Develop the formula  $a_n = a + (n - 1)d$  for the  $n$ th term of an AP, where  $a$  is the first term and  $n \geq 1$ , and use it to solve problems
- Recognise that  $a_n$  is a [linear function](#) of  $n$  in an AP
- Develop the formula  $S_n = \frac{n}{2}(a + a_n)$  for the  $n$ th partial sum of an AP, and use the formula to solve problems
- Develop the formula  $S_n = \frac{n}{2}[2a + (n - 1)d]$  for the  $n$ th partial sum of an AP, and use the formula to solve problems
- Apply the formulas for arithmetic sequences and their partial sums to model and solve growth and decay problems involving a quantity that is a linear function of time

### Geometric sequences and series

- Define a sequence  $a_n$  to be a [geometric sequence](#), or [geometric progression \(GP\)](#), if every ratio  $\frac{a_n}{a_{n-1}}$  of successive terms is the same where  $n \geq 2$ , that is  $a_n = ra_{n-1}$  for some non-zero [real number](#)  $r$  called the [common ratio](#)
- Develop the formula  $a_n = ar^{n-1}$  for the  $n$ th term of a GP, where  $a$  is the first term and  $n \geq 1$ , and use it to solve problems
- Recognise that  $a_n$  is an [exponential function](#) of  $n$  in a GP
- Prove by expansion  $(x - 1)(x^{n-1} + x^{n-2} + \dots + x^2 + x + 1) = x^n - 1$  for whole numbers  $n \geq 1$

- Develop the formula  $S_n = \frac{a(1-r^n)}{1-r}$  for the sum of the first  $n$  terms of a GP where  $r \neq 1$ , and use this formula to solve problems
- Examine the behaviour of  $a_n$  and  $S_n$  as  $n \rightarrow \infty$  for a GP when  $|r| < 1$
- Develop the formula  $S_\infty = \frac{a}{1-r}$  for the limiting sum of a [geometric series](#) with  $|r| < 1$ , and use this formula to solve problems
- Apply the formulas for geometric sequences and series to model and solve growth and decay problems where a quantity is an exponential function of time

# Differential calculus

## Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- selects and applies differentiation methods to solve problems **MAV-12-04**

## Content

### Differentiation with exponential functions

- Apply the rules of [differentiation](#) to find the [derivative](#) of  $f(x) = ke^{ax}$ , where  $k$  and  $a$  are [constants](#)
- Use the [chain rule](#) to prove  $\frac{d}{dx}(e^{ax+b}) = ae^{ax+b}$ , where  $a$  and  $b$  are constants and  $a \neq 0$
- Prove and use the formula  $\frac{d}{dx}(a^x) = (\ln a)a^x$ , where  $a$  is a constant and  $a > 0$
- Use the chain rule to differentiate [functions](#) of the form  $e^{f(x)}$

### Differentiation with logarithmic functions

- Use chain rule on the [identity](#)  $e^{\ln x} = x$  to prove the formula  $\frac{d}{dx}(\ln x) = \frac{1}{x}$  for the derivative of the [natural logarithm](#) function where  $x > 0$
- Prove and use the formula  $\frac{d}{dx}(\log_a x) = \frac{1}{x \ln a}$ , where  $a$  is a constant and  $a > 0$
- Use the chain rule to differentiate functions of the form  $\ln f(x)$

### Differentiation with trigonometric functions

- Determine the formulas  $\frac{d}{dx}(\sin x) = \cos x$  and  $\frac{d}{dx}(\cos x) = -\sin x$  informally by using the [graphs](#) of  $y = \sin x$  and  $y = \cos x$  and verify with graphing applications
- Use the rules of differentiation to show that  $\frac{d}{dx}(\tan x) = \sec^2 x$
- Use the rules of differentiation to find the derivatives of  $\operatorname{cosec} x$ ,  $\sec x$  and  $\cot x$
- Use the chain rule to differentiate functions of the form  $\sin f(x)$ ,  $\cos f(x)$  and  $\tan f(x)$

### Using derivatives

- Apply the product, quotient and chain rules to differentiate functions of the form  $f(x)g(x)$ ,  $\frac{f(x)}{g(x)}$  or  $f(g(x))$ , where  $f(x)$  and  $g(x)$  are any of the functions within the scope of the Mathematics Advanced course
- Solve problems involving [equations](#) of [tangents](#) and [normals to curves](#) involving any of the functions within the scope of the Mathematics Advanced course

# Integral calculus

## Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- solves problems involving indefinite and definite integrals **MAV-12-05**

## Content

### Primitive functions

- Define a [primitive](#) of a [function](#)  $f(x)$  as a function  $F(x)$  whose [derivative](#)  $F'(x) = f(x)$  and recognise the process of finding the primitive as the reverse of [differentiation](#)
- Recognise that a function whose derivative is everywhere [zero](#) is a [constant function](#)
- Prove by differentiation that a primitive of  $f(x) = x^n$  is  $F(x) = \frac{x^{n+1}}{n+1}$ , for all real  $n \neq -1$
- Prove by differentiation that if  $F(x)$  and  $G(x)$  are primitives of  $f(x)$  and  $g(x)$ , and  $k$  is a [constant](#), then  $F(x) + G(x)$  is a primitive of  $f(x) + g(x)$ , and  $kF(x)$  is a primitive of  $kf(x)$
- Recognise that primitives of a function  $f(x)$  are not unique, and that any two primitives of  $f(x)$  differ by a constant, so that if  $F(x)$  is a primitive of  $f(x)$ , the general primitive of  $f(x)$  is  $F(x) + C$ , for some constant  $C$
- Determine the primitive of a given function  $f(x)$ , where  $f(x)$  is a sum of functions of the form  $kx^n$  for all real  $n \neq -1$
- Determine the primitive function for functions of the form  $f(x) = (ax + b)^n$ , for all real  $n \neq -1$ , where  $a$  and  $b$  are constants
- Use algebraic manipulation to express given functions in forms suitable for determining primitive functions
- Determine  $f(x)$ , given  $f'(x)$  and an initial condition  $f(a) = b$  where  $a$  and  $b$  are constants

### The definite integral

- Examine for a function  $f(x)$ , which indicates the [rate of change](#) of a quantity, the meaning of  $\sum_b f(x)\Delta x$ , where the interval  $a \leq x \leq b$  is divided into subintervals of length  $\Delta x$ , and describe  $\sum_a f(x)\Delta x$  as an estimate of the total change in that quantity over the interval  $a \leq x \leq b$
- Consider the definite integral as  $\int_a^b f(x)dx = \lim_{\Delta x \rightarrow 0} \sum_a f(x)\Delta x$ , noting that this implies that the result of a definite integral will be negative when  $f(x) \leq 0$  throughout the interval  $a \leq x \leq b$
- Define informally that a function is continuous on the interval  $a \leq x \leq b$  if it can be drawn between the two endpoints of the interval without taking the pen off the paper
- Graph the region between the [continuous function](#)  $y = f(x)$  and the  $x$ -axis, where  $f(x) \geq 0$  on the interval  $a \leq x \leq b$
- Use a graphing application to compare different methods of approximating the area,  $A$ , of the region between the continuous function  $y = f(x)$  and the  $x$ -axis, where  $f(x) \geq 0$  on the interval  $a \leq x \leq b$ , by summing the areas of trapezia or rectangles each of width  $\Delta x = \frac{b-a}{n}$  and approximate height  $f(x)$  for any  $x$  lying in its base, and observe the effect on the precision of the approximation of  $A$  as the number  $n$  of subintervals of  $a \leq x \leq b$  increases, that is as  $\Delta x \rightarrow 0$

- Evaluate the definite integral  $\int_a^b f(x)dx$  by calculating areas using geometrical formulas, where the shape of  $f(x)$  allows such calculations, in cases where  $f(x) \geq 0$  throughout  $a \leq x \leq b$ ,  $f(x) \leq 0$  throughout  $a \leq x \leq b$  or where  $f(x)$  changes sign in the interval  $a \leq x \leq b$

### The Fundamental Theorem of Calculus

- Consider the function defined by  $A(x) = \int_a^x f(t)dt$  and use a graphing application to recognise that  $A(x)$  is a primitive of  $f(x)$
- Recognise the [Fundamental Theorem of Calculus](#) as  $\int_a^b f(x)dx = [F(x)]_a^b = F(b) - F(a)$  for a continuous function  $f$  on the interval  $a \leq x \leq b$  where  $F(x)$  is any primitive of  $f(x)$

### Indefinite integrals

- Use the notation  $\int f(x)dx$  for the general primitive of  $f(x)$ , called the [indefinite integral](#) of  $f(x)$ , so that  $\int f(x)dx = F(x) + C$ , for some constant  $C$ , where  $F(x)$  is any primitive of  $f(x)$
- Recognise [integration](#) as the process of finding the indefinite integral of a function
- Use the formula  $\int x^n dx = \frac{1}{n+1}x^{n+1} + C$  for real  $n \neq -1$
- Use the identities  $\int (f(x) + g(x))dx = \int f(x)dx + \int g(x)dx$  and  $\int kf(x)dx = k\int f(x)dx$  for primitives
- Prove by differentiation, and apply  $\int u^n \frac{du}{dx} dx = \frac{1}{n+1}u^{n+1} + C$ , where  $u$  is a function of  $x$ , or  $\int f'(x)[f(x)]^n dx = \frac{1}{n+1}[f(x)]^{n+1} + C$ , for real  $n \neq -1$

### Integration with exponential functions

- Establish and use the formula  $\int e^x dx = e^x + C$
- Establish and use the formula  $\int e^{ax+b} dx = \frac{1}{a}e^{ax+b} + C$ , where  $a$  and  $b$  are constants and  $a \neq 0$
- Establish and use the formula  $\int a^x dx = \frac{a^x}{\ln a} + C$ , where  $a$  is a constant and  $a > 0$
- Establish and use  $\int e^u \frac{du}{dx} dx = e^u + C$ , where  $u$  is a function of  $x$ , or  $\int f'(x)e^{f(x)} dx = e^{f(x)} + C$
- Find primitives of functions involving [exponential functions](#)

### Integration with logarithmic functions

- Derive and use the formula  $\int \frac{1}{x} dx = \ln|x| + C$  where  $x \neq 0$
- Establish and use the formula  $\int \frac{1}{ax+b} dx = \frac{1}{a}\ln|ax+b| + C$ , where  $a$  and  $b$  are constants and  $a \neq 0$
- Establish and use  $\int \frac{u'}{u} dx = \ln|u| + C$ , where  $u$  is a function of  $x$ , or  $\int \frac{f'(x)}{f(x)} dx = \ln|f(x)| + C$ , on a [domain](#) where  $f(x) \neq 0$

### Integration with trigonometric functions

- Establish and use the formulas  $\int \sin x dx = -\cos x + C$ ,  $\int \cos x dx = \sin x + C$  and  $\int \sec^2 x dx = \tan x + C$
- Establish and use indefinite integrals of the form  $\int f(ax+b)dx$ , where  $a$  and  $b$  are constants and  $a \neq 0$ , and  $f(x) = \sin x$ ,  $f(x) = \cos x$  and  $f(x) = \sec^2 x$
- Determine indefinite integrals of the form  $\int f'(x)\sin f(x)dx$ ,  $\int f'(x)\cos f(x)dx$  and  $\int f'(x)\sec^2 f(x)dx$

## Areas and the definite integral

- Apply  $\int_a^b f(x)dx = F(b) - F(a)$ , where  $F(x)$  is a primitive of  $f(x)$ , to calculate definite integrals and solve related theoretical problems involving functions within the scope of the Mathematics Advanced course
- Describe, in the case where  $f(x) \geq 0$  for all values of  $x$  in the interval  $a \leq x \leq b$ , the area bounded by the graph of the continuous function  $y = f(x)$ , the  $x$ -axis and the lines  $x = a$  and  $x = b$ , as  $\int_a^b f(x)dx$
- Recognise, in the case where  $f(x) \leq 0$  for all values of  $x$  in the interval  $a \leq x \leq b$ , the area bounded by the graph of the continuous function  $y = f(x)$ , the  $x$ -axis and the lines  $x = a$  and  $x = b$ , as  $\left| \int_a^b f(x)dx \right|$  or  $-\int_a^b f(x)dx$
- Conclude, for a continuous function  $y = f(x)$  on the interval  $a \leq x \leq b$ , that  $\int_a^b f(x)dx =$  (area of regions between curve and  $x$ -axis lying above the  $x$ -axis)  $-$  (area of regions between curve and the  $x$ -axis lying below the  $x$ -axis)
- Use definite integrals to solve problems involving the areas of regions bounded by the graph of the continuous function  $y = f(x)$ , the  $x$ -axis and the lines  $x = a$  and  $x = b$ , in cases where  $f(x) \geq 0$  throughout  $a \leq x \leq b$ ,  $f(x) \leq 0$  throughout  $a \leq x \leq b$  or where  $f(x)$  changes sign in the interval  $a \leq x \leq b$ , with or without the graph provided
- Use definite integrals to solve problems involving the areas of regions bounded by the graph of the continuous function  $y = f(x)$ , the  $y$ -axis and the lines  $y = a$  and  $y = b$ , in cases where  $x \geq 0$  throughout  $a \leq y \leq b$ ,  $x \leq 0$  throughout  $a \leq y \leq b$  or where  $x$  changes sign in the interval  $a \leq y \leq b$  with or without the graph provided
- Use the fact that the graphs of  $y = a^x$  and  $y = \log_a x$  are [reflections](#) of each other in the line  $y = x$  to solve problems involving areas between the  $x$ -axis or  $y$ -axis and a curve involving either an exponential or [logarithmic function](#)
- Recognise and use the result, where  $f(x)$  is continuous on the interval  $a \leq x \leq c$ ,  $\int_a^b f(x)dx + \int_b^c f(x)dx = \int_a^c f(x)dx$  for all  $c$  such that  $a \leq b \leq c$
- Define and use the result  $\int_a^b f(x)dx = -\int_b^a f(x)dx$ , where  $f(x)$  is continuous on the interval  $a \leq x \leq b$
- Recognise and use symmetry, particularly [odd](#) and [even functions](#), to simplify and solve integration problems
- Use the [Trapezoidal rule](#) to approximate integrals
- Use an online computational application to evaluate definite and indefinite integrals involving functions within and beyond the scope of the Mathematics Advanced course
- Model and solve practical problems involving integrals and areas of regions bounded by a curve and the  $x$ -axis, or by a curve and the  $y$ -axis, involving functions within the scope of the Mathematics Advanced course

# Applications of calculus

## Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- applies calculus to graph functions and model and solve problems involving optimisation, rates of change and motion in a line **MAV-12-06**

## Content

### Turning points, inflections and graphing

- Define a [function](#)  $f(x)$  to be [differentiable](#) at  $x = a$  if  $\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$  exists, that is if there is a non-vertical [tangent](#) to the curve at the point  $P(a, f(a))$  and recognise that if  $f(x)$  is differentiable at  $x = a$  then it is continuous at  $x = a$
- Identify any values of  $x$  where a function is continuous, but not differentiable, given either the [equation](#) of the function or its [graph](#)
- Use repeated [differentiation](#) to find [second derivatives](#) of a function  $y = f(x)$ , denoting them by  $f''(x)$  or  $\frac{d^2y}{dx^2}$  or  $y''$
- Analyse the [stationary points](#) of a function by testing values for  $f'(x)$ , then classify the stationary points as local minima or local maxima where [gradients](#) change around the point, or horizontal points of inflection where gradients have the same sign on both sides of the point
- Interpret the second derivative  $y''$  as the gradient function of the first [derivative](#)  $y'$ , and deduce that if  $y'' > 0$  the curve is [concave up](#) and if  $y'' < 0$  the curve is [concave down](#)
- Define a [point of inflection](#) on a curve as a point where the concavity changes
- Analyse the value of  $f''(x)$  either side of the roots of  $f''(x) = 0$ , and use the resulting concavities to identify which [zeroes](#) of  $f''(x)$  are points of inflection
- Use the second derivative to classify a stationary point as a [local maximum](#), [local minimum](#) or a horizontal point of inflection
- Graph a function by determining local maxima and minima and points of inflection, horizontal and non-horizontal, considering any even or [odd](#) symmetry, the [domain](#), any vertical [asymptotes](#) or other [discontinuities](#), and where applicable, the behaviour of a function as  $x \rightarrow \pm \infty$
- Graph  $y = f'(x)$  and  $y = f''(x)$  for a function  $y = f(x)$ , given only a graph of  $y = f(x)$

### Optimisation

- Define a [global maximum](#) of a function  $f(x)$  to be a point  $P(a, f(a))$  on the graph where  $f(x) \leq f(a)$ , for all  $x$  in the domain, and define a [global minimum](#) similarly
- Examine whether any discontinuities or endpoints of the domain on which  $f(x)$  is being considered are points of maxima or minima
- Model optimisation problems in a variety of contexts by defining [variables](#), noting domain restrictions if necessary, and establishing functions to represent the relationship between variables

- Solve optimisation problems by using calculus to find local and global maxima and minima of differentiable functions, checking discontinuities of  $f(x)$  and endpoints of the domain if applicable
- Formulate conclusions to optimisation problems by evaluating solutions given the constraints of the domain

## Rates of change

- Use differentiation to find and interpret the first and second derivatives,  $\frac{dQ}{dt}$  and  $\frac{d^2Q}{dt^2}$ , in practical problems where a quantity  $Q$  is a function of time  $t$
- Use [integration](#) to solve practical problems on the [rate of change](#) of a quantity  $Q$ , where  $\frac{dQ}{dt}$  or  $\frac{d^2Q}{dt^2}$  is given as a function of time  $t$ , together with sufficient initial conditions
- Describe and examine the graphs of practical situations where the rate of change of a quantity is proportional to the quantity
- Explain that if the rate of change of a positive quantity  $Q$  over time  $t$  is proportional to the size of  $Q$ , then this may be represented by  $\frac{dQ}{dt} = kQ$  where  $k$  is the [constant](#) of proportionality, and if  $k > 0$  the quantity  $Q$  is increasing at a rate proportional to the value of  $Q$  at time  $t$ , while if  $k < 0$  the quantity  $Q$  is decreasing at a rate proportional to the value of  $Q$  at time  $t$
- Verify by [substitution](#) that the function  $Q = Ae^{kt}$  satisfies the relationship  $\frac{dQ}{dt} = kQ$  with  $A$  being the initial value of  $Q$
- Recognise that the equation  $\frac{dQ}{dt} = kQ$  where  $Q > 0$ , and its solution  $Q = Ae^{kt}$  represent [exponential growth](#) when  $k > 0$  and exponential decay when  $k < 0$
- Graph the function  $Q = Ae^{kt}$ , where  $k > 0$  and  $k < 0$  and  $A > 0$  and  $A < 0$ , with and without graphing applications, and identify any asymptotes
- Model and solve growth and decay problems in various contexts using  $\frac{dQ}{dt} = kQ$ ,  $Q = Ae^{kt}$  and the graph of  $Q = Ae^{kt}$  for  $t \geq 0$ , and justify conclusions in the context of the problem
- Determine the [velocity](#) and [acceleration](#) of a particle moving in a straight line given its [displacement](#) from a point as a function of time, and use the notation  $\frac{d^2x}{dt^2}$  or  $\ddot{x}$  to represent acceleration
- Solve problems relating to the motion of a particle moving in a straight line, using both differentiation and integration to connect the concepts of displacement, velocity and acceleration

# Random variables

## Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- solves problems involving discrete probability distributions, continuous random variables and the normal distribution **MAV-12-07**

## Content

### Discrete random variables

- Use the relative frequencies of [discrete random variable](#) datasets to estimate the probabilities that the [random variable](#)  $X$  takes each of its values, and explain why these estimated probabilities add to 1
- Denote the [probability](#) that the discrete random variable  $X$  takes the value  $x$  by  $P(X = x)$ , or by  $P(x)$  if the discrete random variable  $X$  is understood
- Define a [discrete probability distribution](#) to be the [set](#) of values  $x$  taken by a discrete random variable  $X$ , together with the probabilities  $P(x)$  that  $x$  is the [outcome](#) of the [experiment](#)
- Define a discrete random variable to be uniformly distributed if it has [finitely](#) many values, all with the same probability, and use it to model random phenomena with [equally likely outcomes](#)
- Recognise the [mean](#), or [expected value](#),  $\mu$  or  $E(X)$ , of a discrete random variable  $X$  as a [measure of centre](#) for its distribution
- Generate and use the formulas  $E(X) = \mu = \sum xP(x)$  and  $\text{Var}(X) = \sigma^2 = \sum (x - \mu)^2 P(x) = \sum x^2 P(x) - \mu^2$  for the random variable  $X$ , where  $\text{Var}(X)$  is the [variance](#) and  $\sigma$  is the [standard deviation](#) of the distribution
- Recognise that the variance is an expected value because  $\text{Var}(X) = E((X - \mu)^2)$
- Generate a probability distribution for a given discrete random variable and represent the [probability distribution](#) in graphical and tabular form
- Solve problems involving probabilities, expectation and variance of discrete random variables

### Continuous random variables

- Estimate the probability that a [continuous random variable](#) falls in some interval using relative frequencies and histograms obtained from data
- Recognise that the probability of a particular value for a continuous random variable is 0 and hence that  $P(a < X < b) = P(a \leq X < b) = P(a < X \leq b) = P(a \leq X \leq b)$  since  $P(X = a) = P(X = b) = 0$  when  $X$  is a continuous random variable
- Define the [cumulative distribution function \(CDF\)](#),  $F(x)$ , as the probability of a random variable,  $X$ , having values less than or [equal](#) to  $x$ , so  $F(x) = P(X \leq x)$  and  $P(a \leq x \leq b) = F(b) - F(a)$
- Recognise that the cumulative distribution function,  $F(x)$ , is non-decreasing for all  $x$  in its [domain](#), and graph cumulative distribution functions, given a formula for  $F(x)$ , with and without graphing applications
- Define a [probability density function](#) (PDF),  $f(x)$ , for a random variable  $X$  with cumulative distribution function  $F(x)$  as  $f(x) = F'(x)$  and recognise that  $P(a < X < b) = \int_a^b f(x) dx$

- Recognise the properties of a probability density function:  $f(x) \geq 0$  for all  $x$  in the domain of  $f(x)$ , and  $\int_a^b f(x) dx = 1$  if the domain of  $f(x)$  is  $[a, b]$ , or  $\int_{-\infty}^{\infty} f(x) dx = 1$  if the domain of  $f(x)$  is all real  $x$
- Apply the properties of a probability density function to solve problems and justify conclusions
- Find the [mode](#) from a given probability density function
- Obtain the cumulative distribution function using the formula  $F(x) = \int_a^x f(t) dt$  where  $f(x)$  is a given probability density function defined on the interval  $[a, b]$
- Determine and use the probability density function  $f(x) = \frac{1}{b-a}$  for a continuous [uniform distribution](#) for a random variable  $X$  taking values in the interval  $[a, b]$
- Use a cumulative distribution function to calculate the [median](#) and [quartiles](#) for a continuous random variable
- Find the probability density function from a given cumulative distribution function
- Generate the [expression](#) for the expected value of a continuous random variable,  $E(X) = \mu = \int_a^b xf(x) dx$ , where the probability density function  $f(x)$  is defined on the interval  $[a, b]$
- Generate the expression for the variance of a continuous random variable,  $\text{Var}(X) = E(X^2) - \mu^2 = \int_a^b x^2 f(x) dx - \mu^2$ , where the probability density function  $f(x)$  is defined on the interval  $[a, b]$
- Evaluate the expected value and the variance of a continuous random variable, where the probability density function  $f(x)$  is defined on the interval  $[a, b]$ , that involve [integration](#) of functions within the scope of the Mathematics Advanced course
- Evaluate the expected value and the variance of a continuous random variable, where the probability density function  $f(x)$  is defined on the interval  $[a, b]$ , that involve integration of functions beyond the scope of the Mathematics Advanced course using an online computational application

### The normal distribution

- Identify the [normal distribution](#) as a continuous probability distribution that is used to model many naturally occurring phenomena
- Identify the graph of the probability density function of a normal distribution, the [normal](#) curve, as an 'ideal' bell-shaped curve, symmetrical about its mean which is equal to its mode and median, and as having most values concentrated about the mean
- Identify contexts that can be approximately modelled by a [normal random variable](#)
- Use the notation  $X \sim N(\mu, \sigma^2)$  to represent a normally distributed random variable that has mean  $\mu$  and standard deviation  $\sigma$
- Represent probabilities associated with the normal distribution by areas of shaded regions under the normal curve, which may extend to  $\pm \infty$
- Apply the [empirical rule](#) to make judgements and solve problems involving probabilities of normally distributed data: that for normal distributions, approximately 68% of data lie within one standard deviation of the mean, approximately 95% within two standard deviations of the mean and approximately 99.7% within three standard deviations of the mean
- Use graphing applications to explore the normal distribution, graph the probability density function  $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$ , verify the empirical rule and graph the cumulative distribution function
- Recognise features of the normal curve, and identify the [global maximum](#) and points of inflection
- Distinguish between a standard normal distribution with mean 0 and standard deviation 1, and the non-standard normal distribution with mean  $\mu$  and standard deviation  $\sigma$

- Define the z [-score](#), or standardised score, by the formula  $z = \frac{x-\mu}{\sigma}$ , where  $\mu$  is the mean and  $\sigma$  is the standard deviation, and  $x$  is an observed value of a random variable
- Interpret the z -score as the number of standard deviations a score lies above or below the mean
- Use z -scores to compare scores from different sets of data and justify conclusions
- Use z -scores to identify probabilities of [events](#) less or more extreme than a given outcome and solve problems using tables for the standard normal distribution
- Solve problems involving finding the mean or standard deviation of a normal random variable given the probability of an event less or more extreme than a given outcome
- Use z -scores to make judgements related to probabilities of certain events or given sets of data assuming an underlying normal distribution

# Financial mathematics

## Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- models and solves problems to make informed decisions about financial situations **MAV-12-08**

## Content

### Reducing balance loans

- Recognise a [reducing balance loan](#) algebraically as a [compound interest](#) loan with periodic [repayments](#)
- Examine the effect of varying the [interest rate](#) and repayment amount on the time taken to repay a loan, with or without digital tools or by using a given [graph](#)
- Solve problems that involve reducing balance loans by calculating the total amount paid, [equal](#) periodic repayments, the amount still owing and the time taken to repay the loan

### Annuities

- Identify an [annuity](#) as either an [investment](#) account with regular, equal [contributions](#) and [interest](#) compounding at the end of each period, or as a single sum investment from which regular equal withdrawals are made
- Define and model the [future value](#) of an annuity as the sum of all the payments, together with the interest they have earned
- Examine the effect of varying the amount initially invested, the value of the periodic payment, the interest rate and the duration of the annuity on the total value of the investment, using digital tools
- Solve problems involving annuities in which payments are made at the end of each time period and the future value of an annuity is calculated at the end of one of these periods using the formula for the sum of the first  $n$  terms of a geometric [sequence](#)